CSE299 - WEEK 5 PROGRESS REPORT

Dhaka Air - Air Quality Prediction System

Track 3: ML Model & Dataset Development | Section 14 | Group 8

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Project Status: Dataset preparation completed; baseline and first ML model implemented for initial evaluation.

1. Week 5 Goal

According to the project schedule, the Week 5 milestone is to **build a baseline model and the first machine-learning model**. This report records the work completed toward that milestone using the CAMS-BD air-quality dataset.

2. Dataset Preparation Completed

- Started from the original CAMS-BD raw Excel files covering 2012-2024.
- Identified and extracted Dhaka monitoring-station records.
- Standardized changing column names and column order across yearly files.
- Converted date/time fields to a consistent format and sorted records chronologically.
- Handled duplicate rows, unusable timestamps, missing values, and clearly invalid readings.
- Selected the main pollution and weather variables required for ML experiments.

3. Current Dataset Summary

Item	Current Value
Raw CAMS-BD records checked	1,373,986
Dhaka records extracted	230,883
Cleaned Dhaka records	206,275
Main variables	PM2.5, PM10, temperature, humidity, wind, gases, date/time, location

4. Baseline Model and First ML Model

After cleaning the dataset, we created an initial baseline and trained our first ML model. This is an **early experiment** for Week 5, not the final model-selection result.

Model	MAE	RMSE	R^2
Simple Baseline	54.29	70.90	-
Random Forest	28.00	45.51	0.585

Initial observation: the Random Forest produced lower MAE and RMSE than the simple baseline, so the first ML experiment shows improvement over the reference method. More models and deeper error analysis will be completed in the next stage.

5. Evaluation Metrics Used

MAE (Mean Absolute Error): Average absolute difference between predicted and actual values. Lower is better.

RMSE (Root Mean Squared Error): Measures prediction error while giving more penalty to large mistakes. Lower is better.

R^2 (R-squared): Shows how much variation in the target is explained by the model. Higher is generally better.

6. Current Project Pipeline

Raw CAMS-BD Data → Dhaka Extraction → Cleaning & Preprocessing → Baseline → Random Forest → Initial Evaluation

7. Work Completed vs Next Step

Completed	Next
Raw data inspection and Dhaka extraction	Compare additional ML models
Cleaning and preprocessing	Perform detailed error analysis
Feature selection	Refine the prediction target/features if needed
Simple baseline model	Prepare model-comparison results
First Random Forest model	Begin web/dashboard integration later in schedule

8. Week 5 Summary

The Week 5 milestone has been reached: the raw dataset has been prepared, a simple baseline has been established, and the first Random Forest model has been trained and evaluated. The next development stage will focus on model comparison and error analysis before moving to later web-application features.